Experiment No. 5: Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)

Aim

To identify the hardware interfaces of a Network Hub and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology to observe broadcast behavior in Cisco Packet Tracer.

Theory

Task: Answer the following questions.

What is a hub?

A hub is a basic networking device used to connect multiple computers or devices in a Local Area Network (LAN). It allows devices to communicate with each other by sending data between them.

Explain the working principle of a hub.

A hub works on the physical layer (Layer 1) of the OSI model.

- When a device sends data to the hub,
- The hub receives the data and broadcasts (sends) it to all connected devices,
- Every device checks the data, but only the intended recipient accepts it.

The hub does not filter or direct data intelligently it simply forwards everything to all ports.

What is MAC address?

A MAC (Media Access Control) address is a unique hardware address assigned to a network device (like a computer or smartphone).

- It is used to identify devices within a local network.
- It is usually written in hexadecimal format.
- Example: 00:1A:2B:3C:4D:5E

What is IP address?

An IP (Internet Protocol) address is a unique address assigned to a device connected to a network.

- It identifies a device on a network or the internet.
- It helps in sending and receiving data between devices.

Example of IPv4 address: 192.168.1.1

What is a Local Area Network (LAN)?

A Local Area Network (LAN) is a network that connects computers and devices within a small geographical area such as:

- A home
- A school
- An office building

LANs allow devices to share files, printers, and internet connections.

What is a star topology?

A **star topology** is a network layout where:

- All devices are connected to a **central device** (such as a hub or switch).
- Communication passes through the central device.

Advantages:

- Easy to install and manage
- Failure of one cable does not affect the whole network

If the central device fails, the entire network stops working.

Hardware Components Required

- **PC-PT (x3):** End devices acting as network nodes.
- **Hub-PT:** A generic Physical Layer device used to interconnect the nodes.
- **Copper Straight-Through Cables:** The transmission media used for PC-to-Hub connectivity.

Hardware and Connection Setup

- **Network ID:** 192.168.1.0
- **Subnet Mask:** 255.255.255.0
- **PC Naming Convention:**
    - PC 1: **Batch_Roll_A** (e.g., S11_01_A)
    - PC 2: **Batch_Roll_B** (e.g., S11_01_B)
    - PC 3: **Batch_Roll_C** (e.g., S11_01_C)

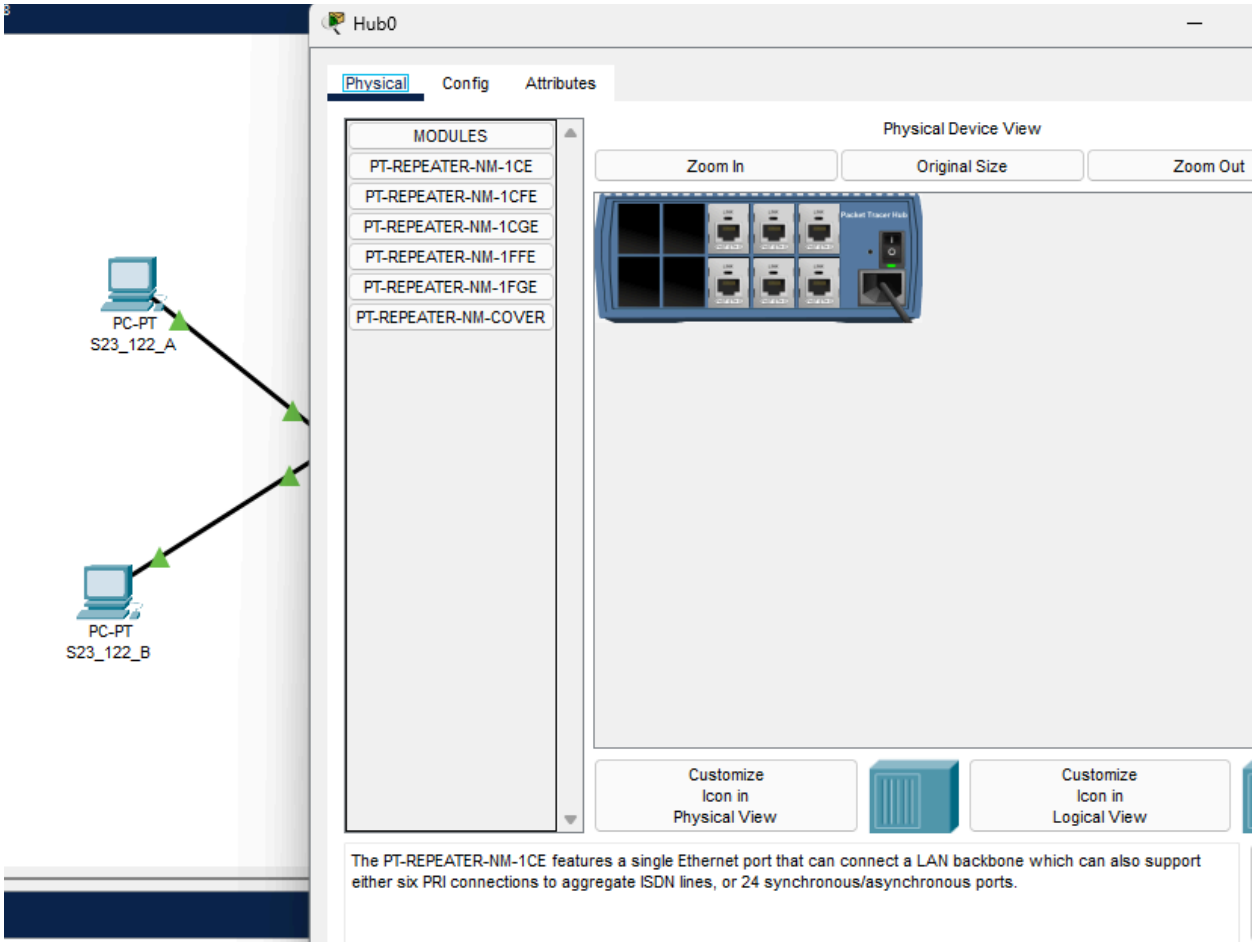
Procedure

Step 1: Physical Topology Design

1. Open Cisco Packet Tracer and select **Network Devices > Hubs > Hub-PT**. Place it in the center.
2. Select **End Devices > PC** and place four units around the Hub.
3. **Naming:** Click on each PC label and rename them following the convention: S11_01_A, S11_01_B, S11_01_C and S11_01_D.

Step 2: Hardware Identification

1. Click on the **Hub** and select the **Physical** tab.
2. Observe the ports. Note that unlike the Switch, a Hub usually has fewer ports and no "intelligence" indicators (like console ports).



Step 3: Physical Cabling

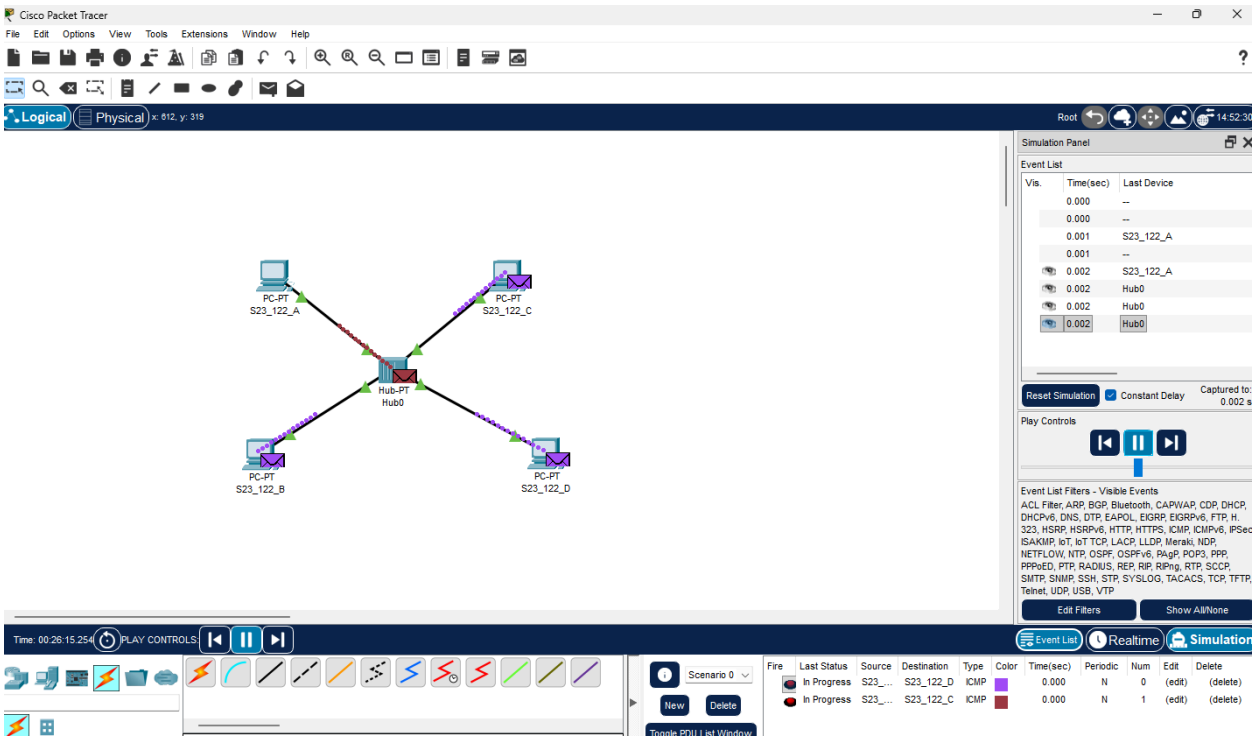
1. Select the **Copper Straight-Through** cable (solid black line).
2. Connect **S11_01_A (FastEthernet 0)** to **Hub (Port 0)**.
3. Repeat for the other three PCs using Hub ports **1, 2** and **3**.
4. **Observation:** Notice that the link lights turn **Green immediately**. Unlike a Switch, a Hub does not run STP (Spanning Tree Protocol), so there is no "Orange" waiting period.

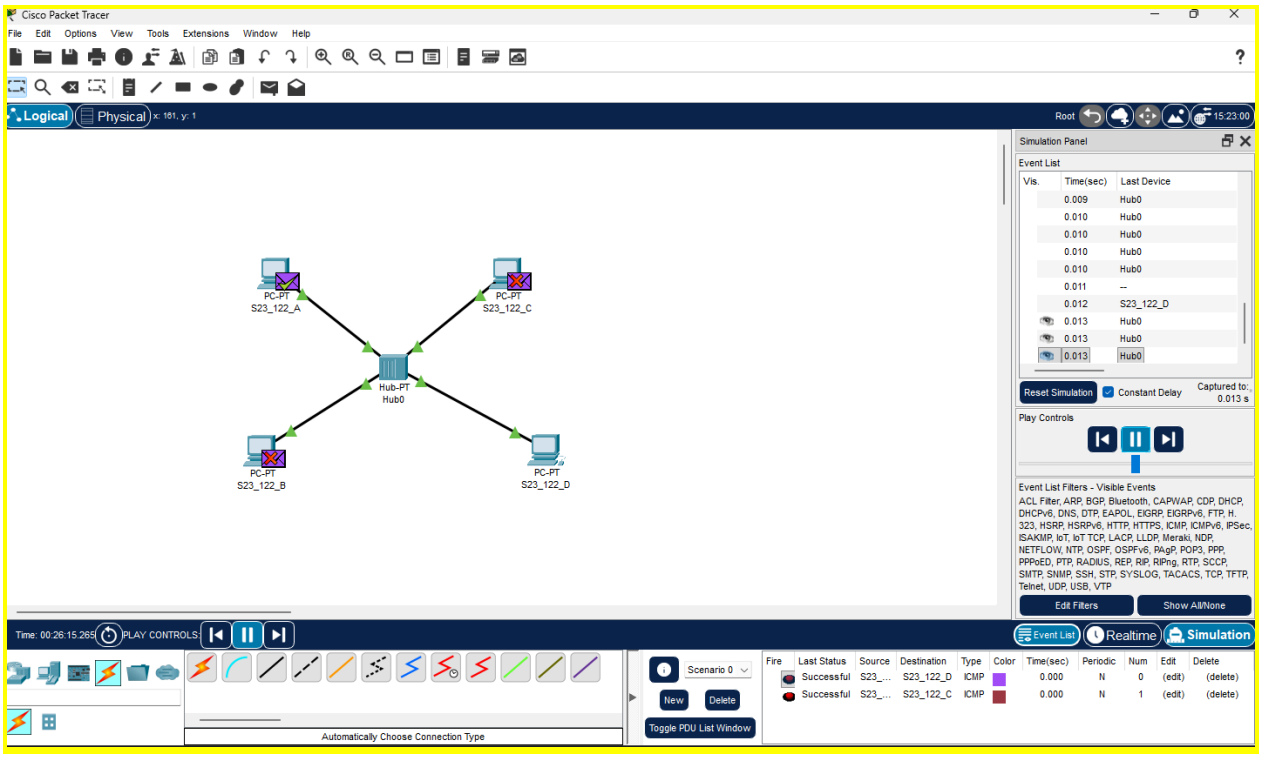
Step 4: Logical Addressing (IP Configuration)

1. Click **S11_01_A > Desktop > IP Configuration**.
2. Enter Static IP: 192.168.1.1 | Subnet: 255.255.255.0.
3. Configure **S11_01_B** with IP 192.168.1.2, **S11_01_C** with IP 192.168.1.3 and **S11_01_D** with IP 192.168.1.4.

Step 5: Testing and Simulation (Crucial Step)

1. **Real-time:** Open the Command Prompt on **S11_01_A** and type ping 192.168.1.3.
2. **Simulation Mode:**
    - Click the **Simulation** tab and filter for **ICMP**.
    - Use the **Add Simple PDU** (Envelope) to click **S11_01_A** then **S11_01_C**.
    - Click **Play** and observe carefully: The packet goes to the Hub, and the Hub sends it to **all** PC B, PC C and PC D.
    - Observe the **Red Cross** on PC B and PC D (because they reject a packet not meant for them) and the **Green Check** on PC C.





6. Observations

- **Instant Connectivity:** Hub ports do not have a "learning" phase; they transition to green immediately upon connection.
- **Broadcast Behavior:** During simulation, the Hub was observed to forward the packet to every connected device. This wastes bandwidth and creates security risks.
- **Device Rejection:** End devices (PCs) are responsible for checking the destination MAC address. If it doesn't match, the PC discards the packet (indicated by the red cross in simulation).

Conclusion

The experiment successfully demonstrates the hardware and functional limitations of a Hub. It concludes that a **Hub** is a non-intelligent Layer 1 device that relies on broadcasting, making it less efficient than a Switch for modern Local Area Networks.